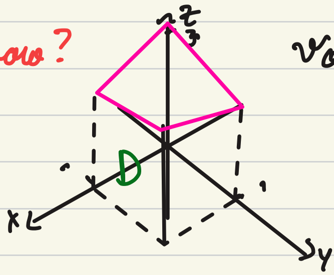
Triple integrals

April 6, 2026

motivation: Let's find the volume of the region in $\mathbb{R}^3$ bounded by the surfaces:

$$z=0\,,\quad x=0\,,\quad y=0\,,\quad x=1\,,\quad y=1\,,\quad z=3-x-y$$

how?



$\|$ Volume under $z=3-x-y$ over the rectangle $[0,1]\times[0,1]=R$

as an iterated double integral

$$V=\iint_R 3-x-y\;dA$$

$$=\int_0^1\int_0^1 3-x-y\;dx\,dy$$

as a triple integral $$=\int_0^1\int_0^1\left[\int_0^{3-x-y} 1\;dz\right]dx\,dy$$

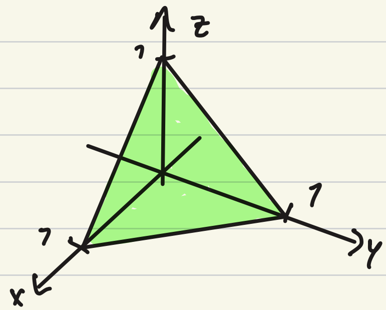
$$=\int_0^1\int_0^1\left(z\Big|_{z=0}^{z=3-x-y}\right)dx\,dy$$

If we want to integrate any continuous function f on D (the solid region in $\mathbb{R}^3$ described above)

we do $$\int_0^1\int_0^1\int_0^{3-x-y} f(x,y,z)\,dz\,dx\,dy$$

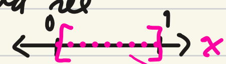
e.g. What is the region

$$D = \{ (x,y,z) \in \mathbb{R}^3 : 0 \le x \le 1, \\ 0 \le y \le 1-x,\ 0 \le z \le 1-y-x \}$$

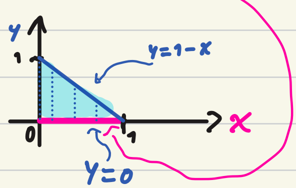


How do we visualize this ?

1) x is "free" to range over $[0,1]$ independently of what happens in y or z.
If we had vision only of what happens in x we would see

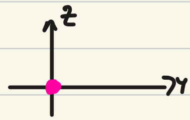


2) The domain for y only depends on x, not on z
If we ignore z, and only look at x and y, we see
for each value of x (in $[0,1]$) $0 \le y \le 1-x$

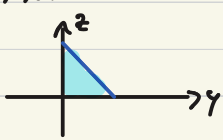


3) The possible values of z depend on x and y.
On top of each point (x,y) in the x-y plane
we let z move between 0 and $1-y-x$ ...
We can take some cross-sections to see how this looks

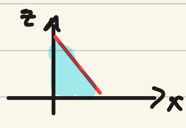
- What happens at $x=1$: obtain a single point $(1,0,0)$



- What happens along $x=0$:
get $0 \le z \le 1-y$



- What happens along $y=0$:
get $0 \le z \le 1-x$



This is how we describe elementary regions in $\mathbb{R}^3$
This region is z-simple, and when projected on the x-y plane is y-simple

Simplest region are "boxes"

in $\mathbb{R}^3 \rightsquigarrow D = \{(x,y,z) : a \le x \le b,\ c \le y \le d,\ \ell \le z \le m\}$

$D = [a,b] \times [c,d] \times [\ell, m]$

Elementary region in $\mathbb{R}^3$:

a description of an elementary region has a "ranking of independence":

1^{st} variable has domain a constant interval $\longrightarrow$ $0 \le y \le 1$

2^{nd} variable has domain an interval with endpoints depending on 1^{st} variable $\longrightarrow$ eg: $0 \le x \le \sqrt{1-y^2}$

3^{rd} variable has domain an interval with endpoints depending on 1^{st} and 2^{nd} variable $\longrightarrow$ $0 \le z \le \sqrt{1-x^2-y^2}$

eg:
$0 \le y \le 1$
$0 \le x \le \sqrt{1-y^2}$
$0 \le z \le \sqrt{1-x^2-y^2}$

Draw cross section $y = 0$, $y = 1/\sqrt{2}$ and then sketch the region

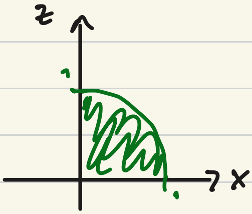
Octant

cross section $y = 0$

$0 \le x \le \sqrt{1} = 1$

$0 \le z \le \sqrt{1-x^2}$

$x^2 + z^2 \le 1$



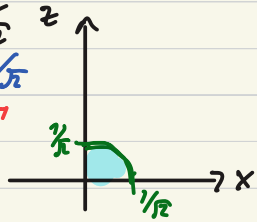
cross section $y = \frac{1}{\sqrt{2}}$

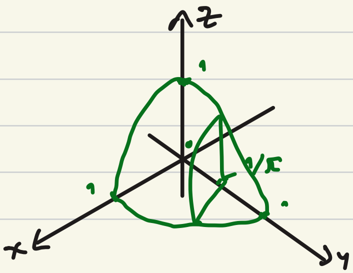
$0 \le x \le \sqrt{1 - 1/2} = 1/\sqrt{2}$

$0 \le z \le \sqrt{1 - x^2 - 1/2}$

$0 \le z \le \sqrt{\frac{1}{2} - x^2}$

$x^2 + z^2 \le 1/2$



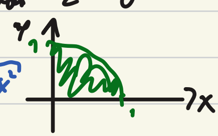


what happens on the x-y plane?

cross section $z = 0$

$0 \le y \le 1$

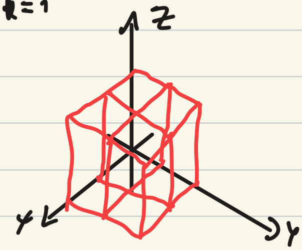
$0 \le x \le \sqrt{1-x^2}$



Integration in $\mathbb{R}^3$

Rigorous definition: If $f : [a,b]\times[c,d]\times[l,n] \to \mathbb{R}$ is a continuous function ($[a,b]\times[c,d]\times[l,n] \subseteq \mathbb{R}^3$), then we define the triple integral of f over $R=[a,b]\times[c,d]\times[l,n]$ (and denote it $\iiint_R f\,dV$) as the limit of Riemann sums

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0\\ \Delta z\to 0}} \sum_{i=1}^{s}\sum_{j=1}^{r}\sum_{k=1}^{t} f(x_i, y_j, z_k)\,\Delta x_i\,\Delta y_j\,\Delta z_k$$

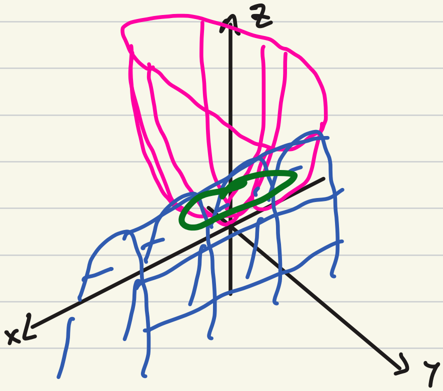


Theorem (Fubini's Theorem): If the region $R \subseteq \mathbb{R}^3$ is a "box" ($R=[a,b]\times[c,d]\times[l,n]$) and the function $f : R\subseteq\mathbb{R}^3 \to \mathbb{R}$ is continuous, then f is Riemann integrable (i.e. the limit definition of the integral exists) and we have the equality:

$$\iiint_R f\,dV = \int_a^b \int_c^d \int_l^n f(x,y,z)\,dz\,dy\,dx$$
$$= \int_c^d \int_a^b \int_l^n f \quad dz\,dx\,dy$$
$$= \dots \qquad \text{all possible orders}$$

Proposition: If $R=\{(x,y,z)\in\mathbb{R}^3 : a\le x\le b,\ \gamma(x)\le y\le\delta(x),\ \mu(x,y)\le z\le\nu(x,y)\}$ is an elementary region and $f: R\to\mathbb{R}$ is cont., then $\iiint_R f\,dV = \int_a^b\left[\int_{\gamma(x)}^{\delta(x)}\left[\int_{\mu(x,y)}^{\nu(x,y)} dz\right]dy\right]dx$

e.g. Find the volume of the region $R \subseteq \mathbb{R}^3$ bounded by $z = 4x^2 + y^2$ and $y^2 + z = 2$



$z = 4x^2+y^2$ ⇓ paraboloid but elongated along y

$y^2+z=2$ ⇓ $z = 2 - y^2$ downward parabola in the y-z plane and extends along x

Let's find points of intersection of the two surfaces

$$z = 4x^2+y^2 = 2-y^2 \rightsquigarrow 4x^2 = 2-2y^2$$
$$\rightsquigarrow 2x^2 = 1 - y^2$$

The intersection is an ellipse $2x^2+y^2=1$